

Inference-Time Compute

Why every token reads the whole model, which model should answer, and the consequences of thinking longer

Required

Chen, Zaharia and Zou, FrugalGPT · [arXiv:2305.05176](#)

Optional

Muennighoff et al., s1 · [arXiv:2501.19393](#) Ong et al., RouteLLM · [arXiv:2406.18665](#)

Wei et al., Chain-of-Thought Prompting · [arXiv:2201.11903](#)

Leviathan et al., Speculative Decoding · [arXiv:2211.17192](#) Cai et al., Medusa · [arXiv:2401.10774](#)

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Quick overview

What decode moves. One token at a time, the whole model read for each.

Measured on your Assignment 1 model, and the one trick that gets more tokens per read

Routers and cascades. Send the easy prompts to the small model.

The router game: you route five prompts before the answers show

Thinking. Chain of thought, from a prompting trick to a trained span with a length limit.

What the extra tokens gain in accuracy and take in time

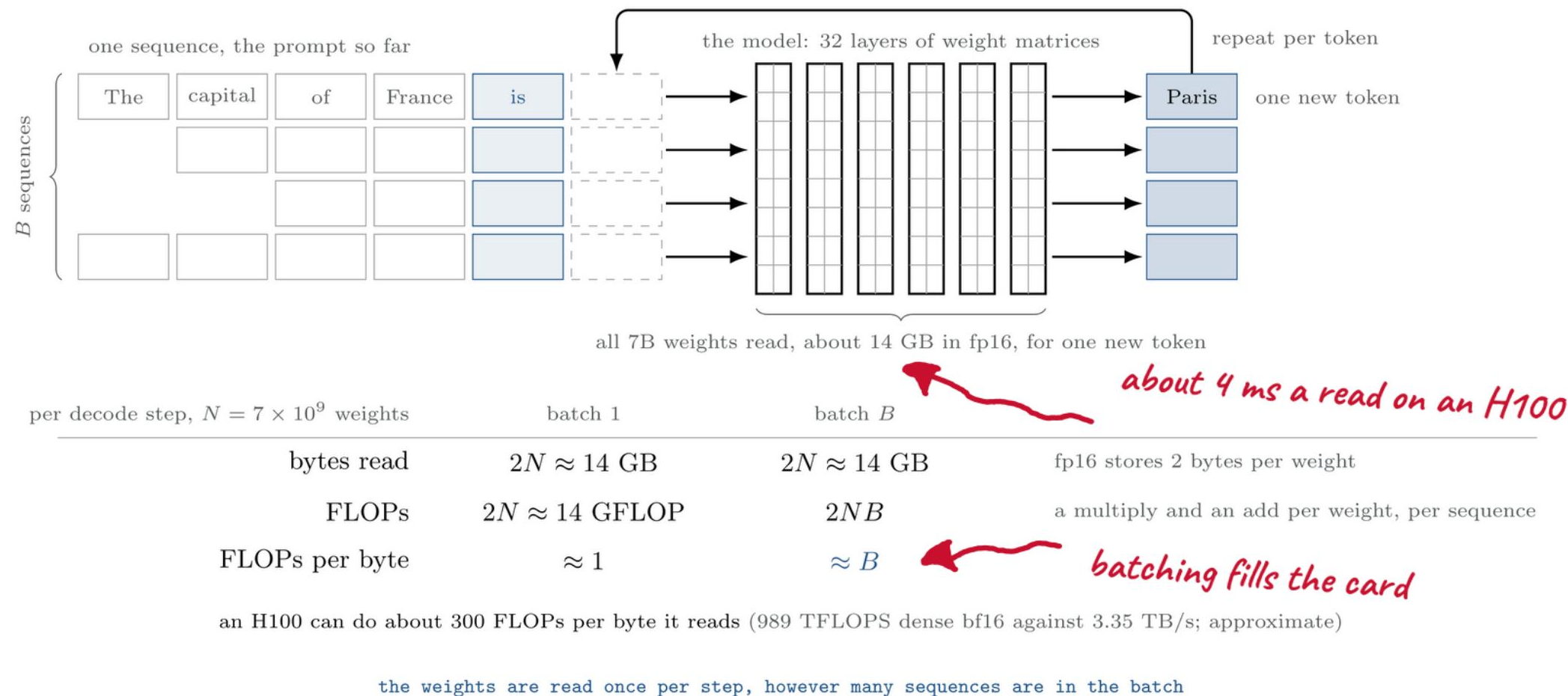
Assignment 5. LoRA on a real checkpoint, out today.

It runs on Thursday's memory arithmetic

1 WHAT DECODE MOVES

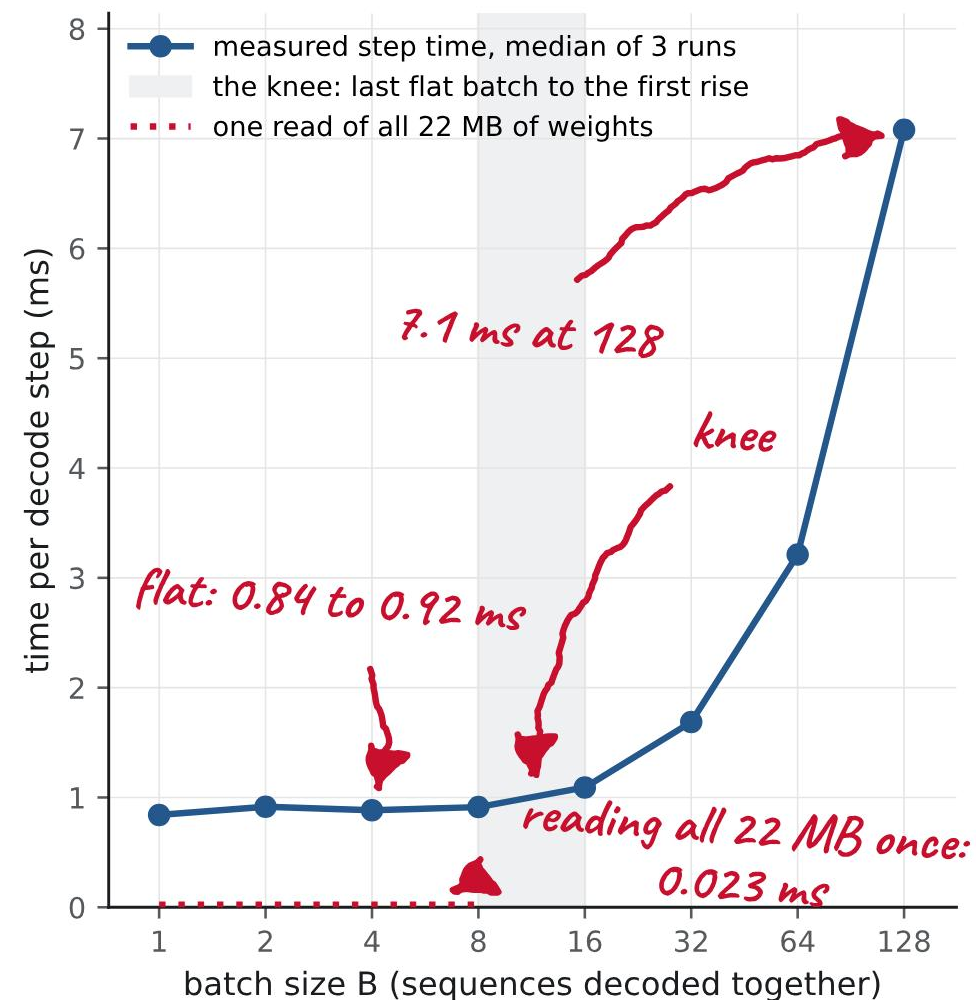
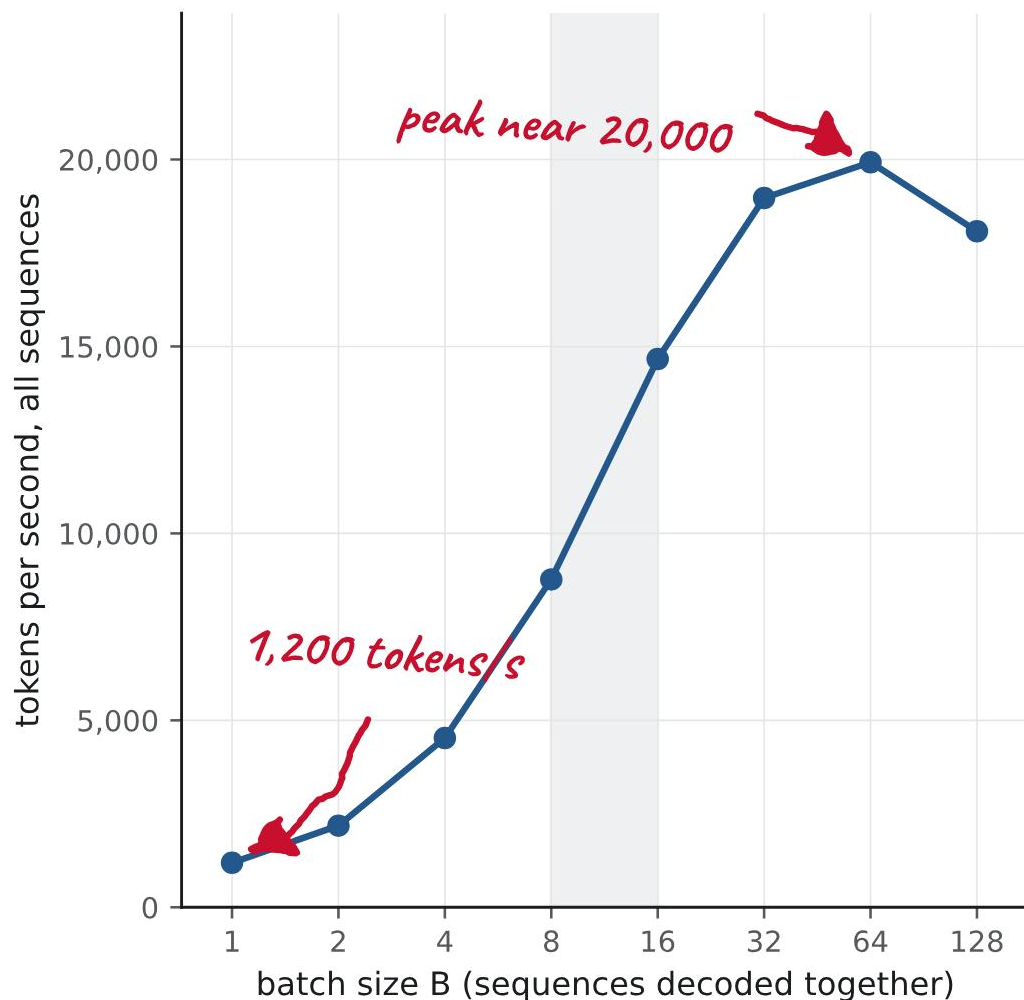
One Token at a Time

What a Step Moves



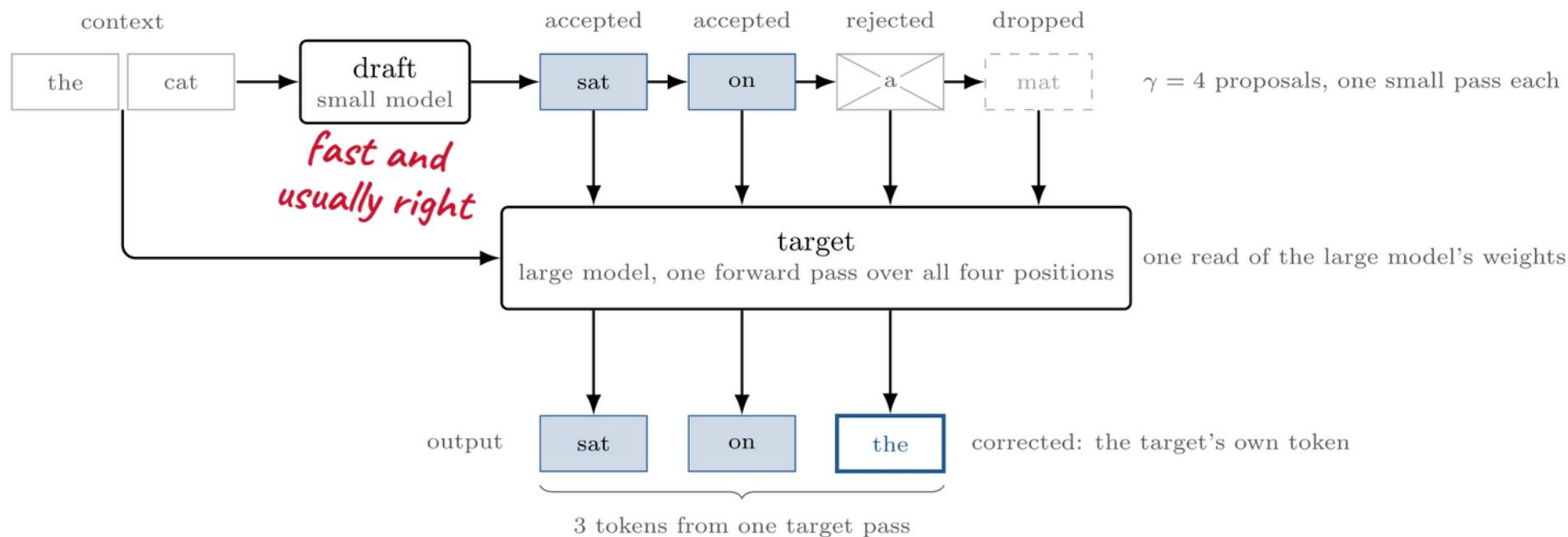
Schematic for a 7B model in fp16. H100 figures from NVIDIA's data sheet (989 TFLOPS dense bf16, 3.35 TB/s HBM3), approximate. Engine-side batching and the KV cache are Mark Zhao's sessions, October 15 to 22.

Time per Step



Draft and Verify

Leviathan, Kalman and Matias, 2023



$$\text{expected tokens per target pass} = \frac{1 - \alpha^{\gamma+1}}{1 - \alpha}$$

α : per-token acceptance rate

at $\alpha = 0.8$ and $\gamma = 4$: about 3.4

÷ the draft's own time

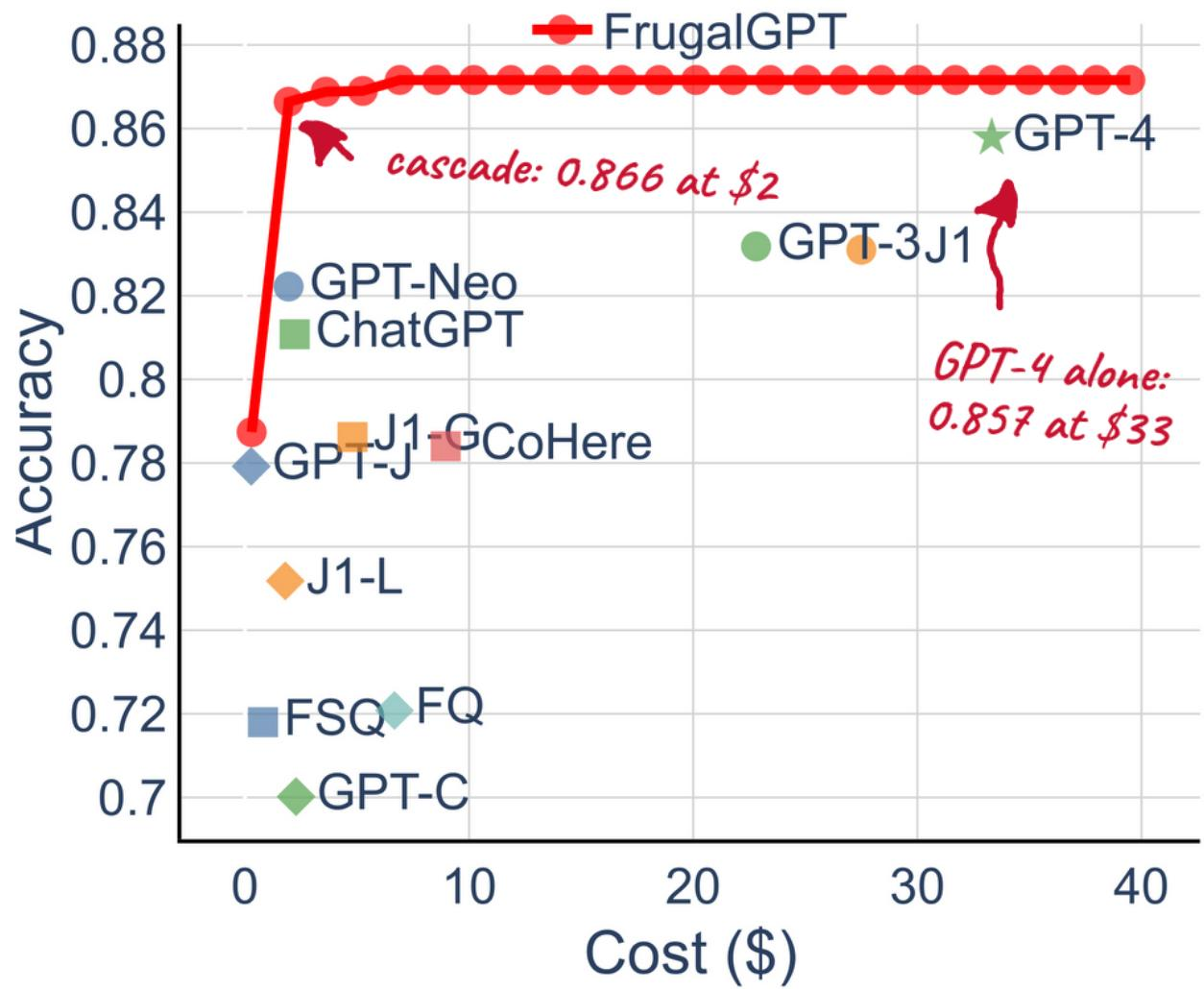
the output distribution is unchanged; only the time per token changes

2

ROUTERS AND CASCADES

Small Model First

Cascades

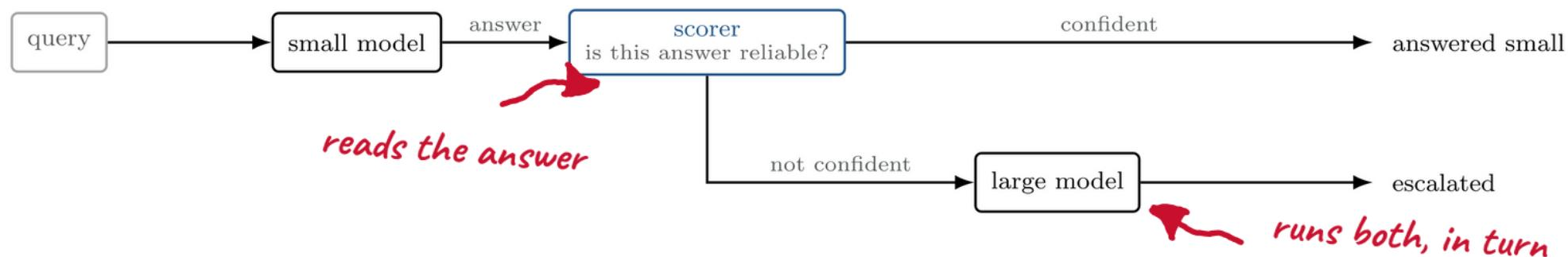


Chen, Zaharia and Zou, "FrugalGPT: How to Use Large Language Models While Reducing Cost and Improving Performance," 2023, arXiv:2305.05176, Figure 1(c) and Table 3. Prices are the paper's 2023 API prices over the HEADLINES evaluation set.

Cascade or Router

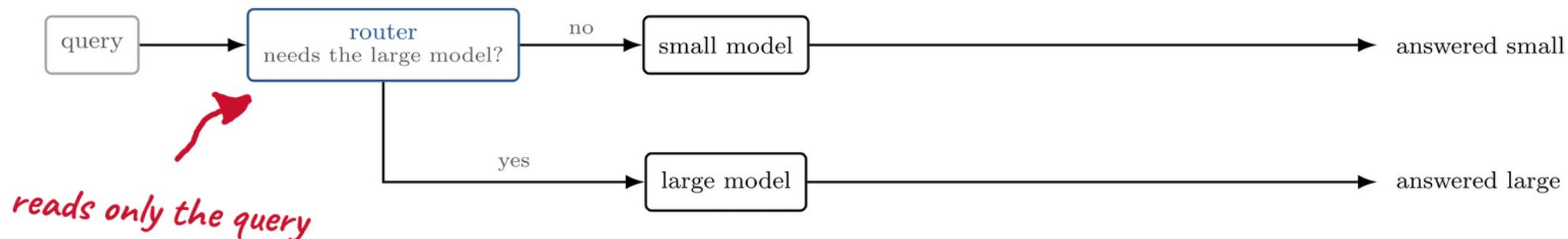
cascade

FrugalGPT: Chen, Zaharia and Zou, 2023



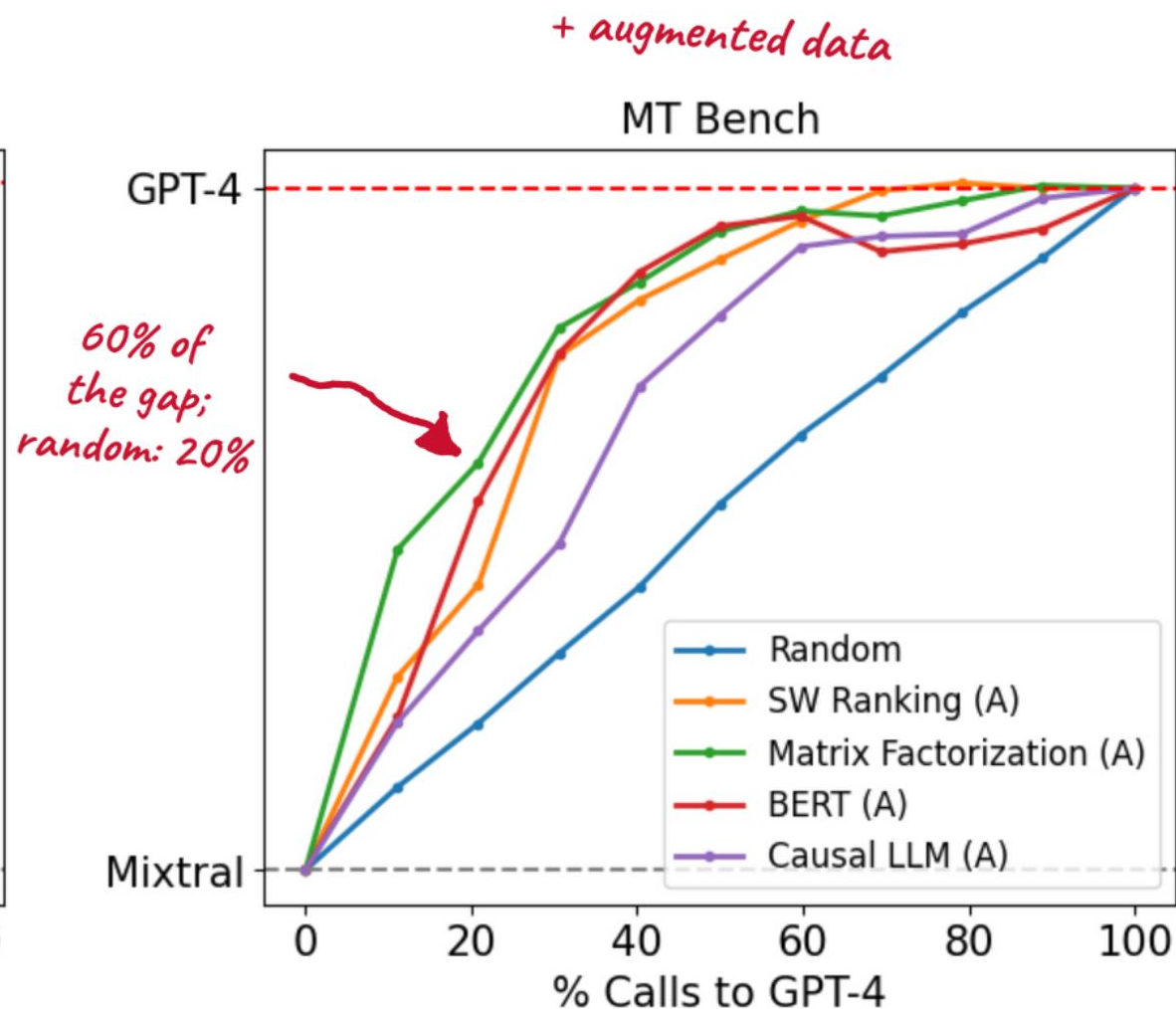
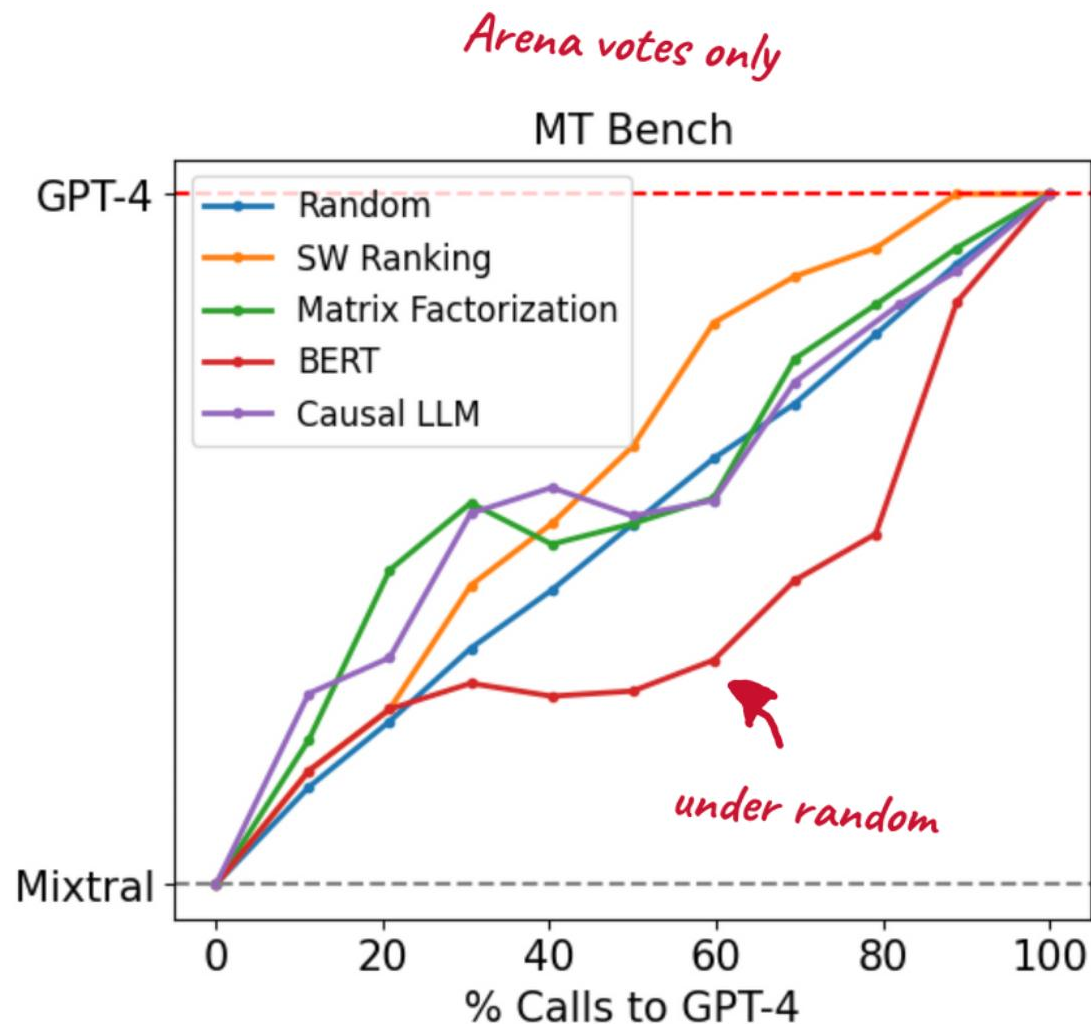
router

RouteLLM: Ong et al., 2024



a cascade decides after the small answer; a router decides before it

Learned Routers



Route Five Prompts

prompt	small (8B)	frontier	price per call (USD)
	Qwen3 8B, thinking off, one RTX 4090	Claude Sonnet	
1 What is the capital of Australia? Answer in one word.	✓ 0.15 s 3 tokens	✓ 2.2 s 6 tokens	0.030
2 Convert to JSON with keys name, age, city: Maria Lopez, 34, lives in Denver. Reply with JSON only.	✓ 0.27 s 21 tokens	✓ 1.9 s 30 tokens	0.025
3 A train leaves at 9:40 and arrives at 13:15. How many minutes is the trip? End with the number.	✓ 0.87 s 180 tokens	✓ 2.6 s 91 tokens	0.025
4 Find the least positive integer n for which 1000 divides $2^n + 5^n - n$. End with n .	✗ 59 s 8200 tokens	✓ 14 s 1900 tokens	0.057
5 Write Python <code>add_months(d, n)</code> for a date; Jan 31 plus one month must give Feb 28 or 29.	✗ 3.3 s 510 tokens	✓ 3.7 s 340 tokens	0.028

My runs, October 2, 2026: five prompts sent verbatim with no system prompt, scored by exact match or a stated check with no judge model. The small model through Ollama on my RTX 4090; the frontier model through the Claude Code CLI. Qwen2.5 7B Instruct, the 2024 model, also scored three of five.

Q ASSIGNMENT 4 QUIZ

Fifteen Minutes

3

REASONING MODELS

Tokens Before the Answer

Chain of Thought

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

*the steps are
plain tokens*

each one a decode step

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

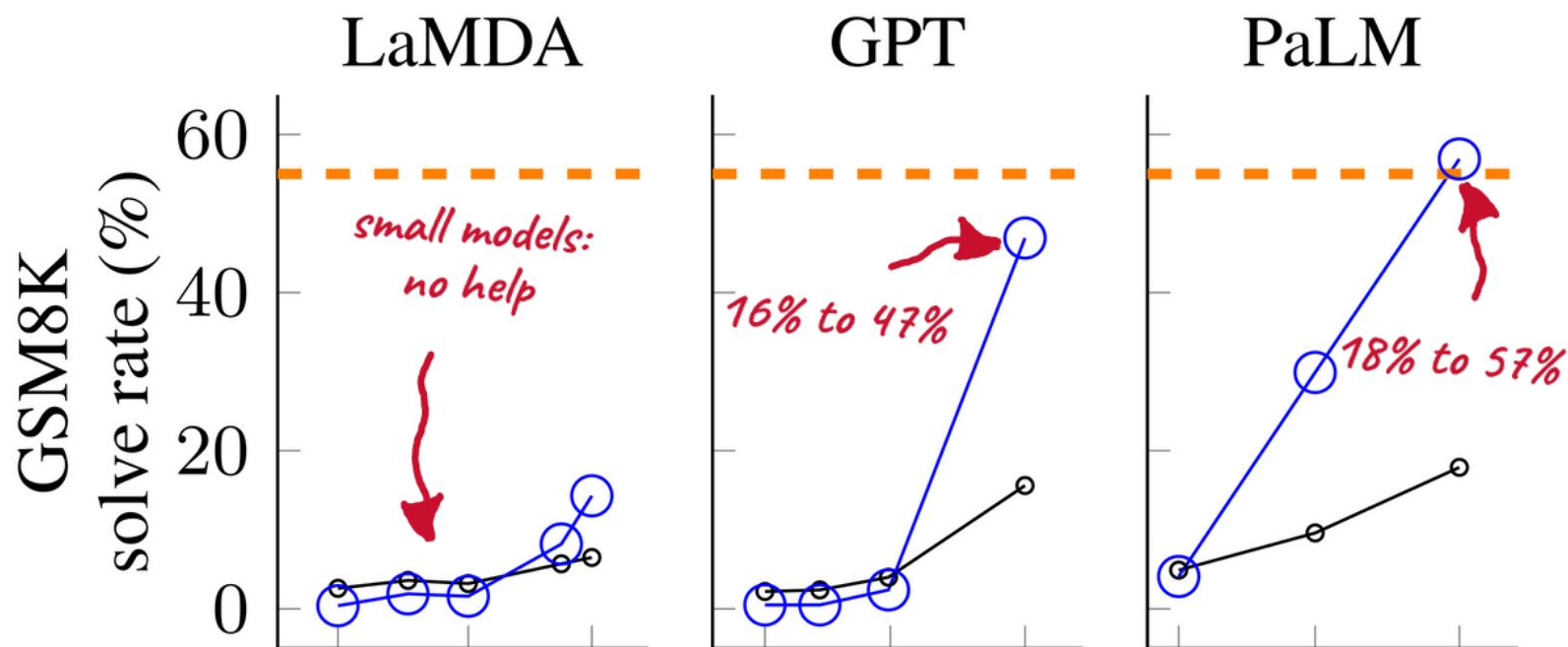
Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✓

Steps and Scale

- Standard prompting
- Chain-of-thought prompting
- - - Prior supervised best



Thinking On and Off

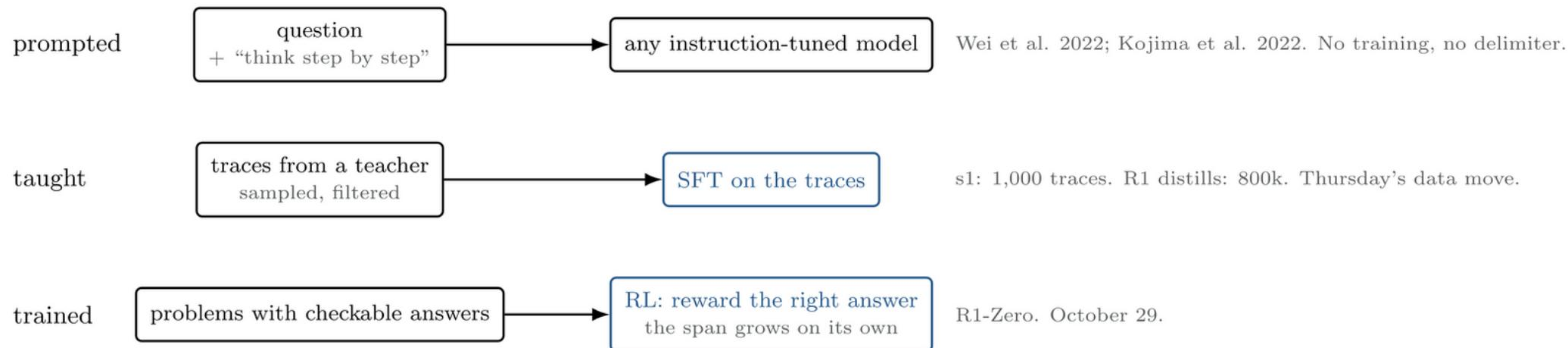
prompt	thinking off	thinking on
	Qwen3 8B, one RTX 4090	the same model and prompts
1 What is the capital of Australia? Answer in one word.	✓ 0.15 s 3 tokens	✓ 1.3 s 180 tokens
2 Convert to JSON with keys name, age, city: Maria Lopez, 34, lives in Denver. Reply with JSON only.	✓ 0.27 s 21 tokens	✓ 1.7 s 230 tokens
3 A train leaves at 9:40 and arrives at 13:15. How many minutes is the trip? End with the number.	✓ 0.87 s 180 tokens	✓ 7.1 s 1100 tokens
4 Find the least positive integer n for which 1000 divides $2^n + 5^n - n$. End with n .	✗ 59 s 8200 tokens	✓ 76 s 10000 tokens
5 Write Python <code>add_months(d, n)</code> for a date; Jan 31 plus one month must give Feb 28 or 29.	✗ 3.3 s 510 tokens	✓ 53 s 7400 tokens

180 tokens for Canberra

both misses now right

My runs, October 2, 2026: Qwen3 8B through Ollama on my RTX 4090, the five router-game prompts, temperature 0, thinking off and on; one run each with thinking on. The quoted span is the model's thinking on the capital prompt, captured in full.

Prompted, Taught, Trained



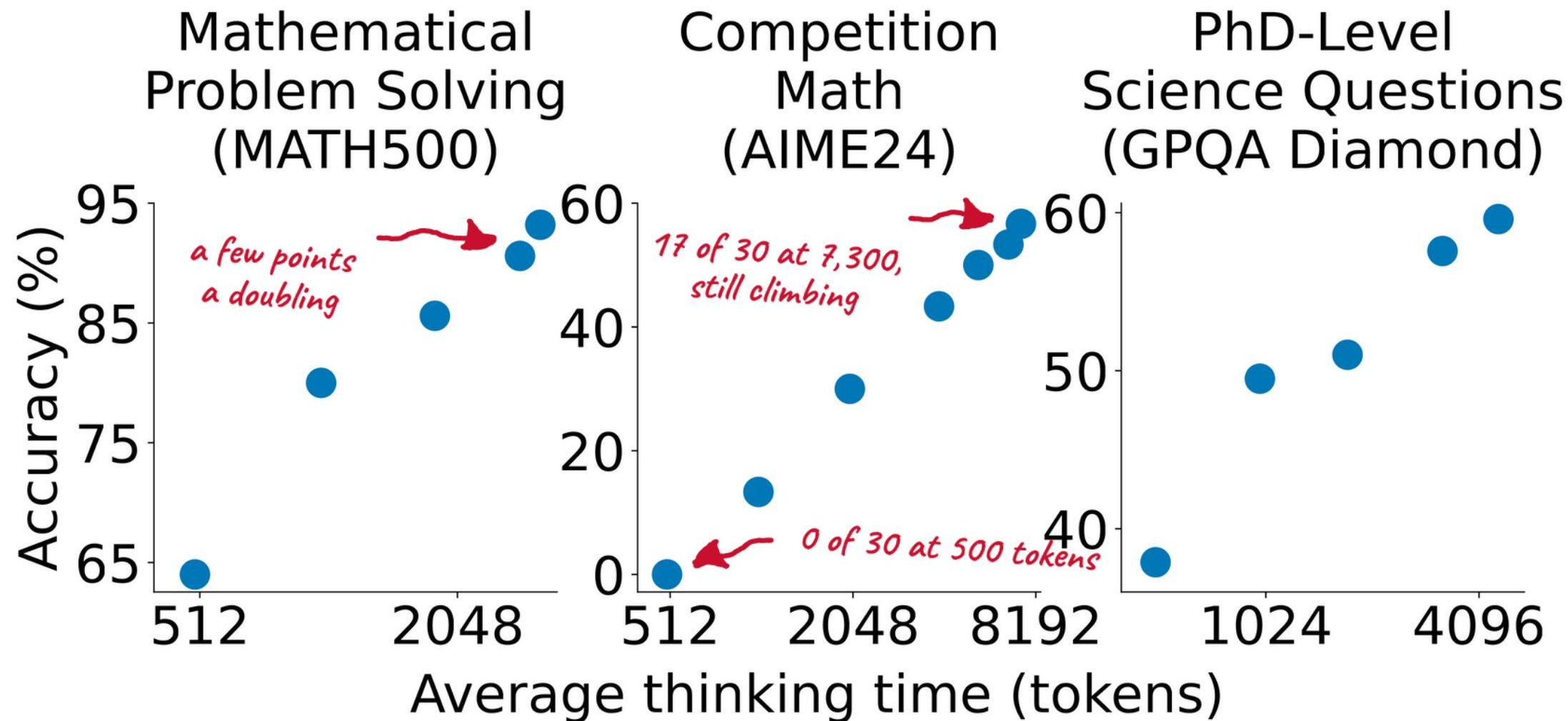
taught and trained spans carry their own end token, which is the handle a length limit grabs



the thinking span: tokens the model writes before it answers, each one a decode step

a token: suppress or insert it

Test-Time Scaling



Budget Forcing

How many r in raspberry?

Question

Let's break down the process of counting the letter 'r' in the word "raspberry" ...

- * First letter: 'r' - This is an 'r', count = 1.
- * Second letter: 'a' - Not an 'r', count remains 1 ...
- * Sixth letter: 'e' - Not an 'r', count remains 1.
- * Seventh letter: 'r' - This is an 'r', count = 2.
- * Eighth letter: 'y' - Not an 'r', count remains 2 ...

The number of 'r's in "raspberry" is 2.

Wait, let's re-read the question carefully. It asks "How many r in raspberry?" ... * r - a - s - p - b - e - r - r - y ... * First 'r' ... * Second 'r' ... * Third 'r' ... Count = 3 ...

Reasoning trace

My initial answer of 2 was incorrect due to a quick reading of the word. **Final Answer:** The final answer is **3**

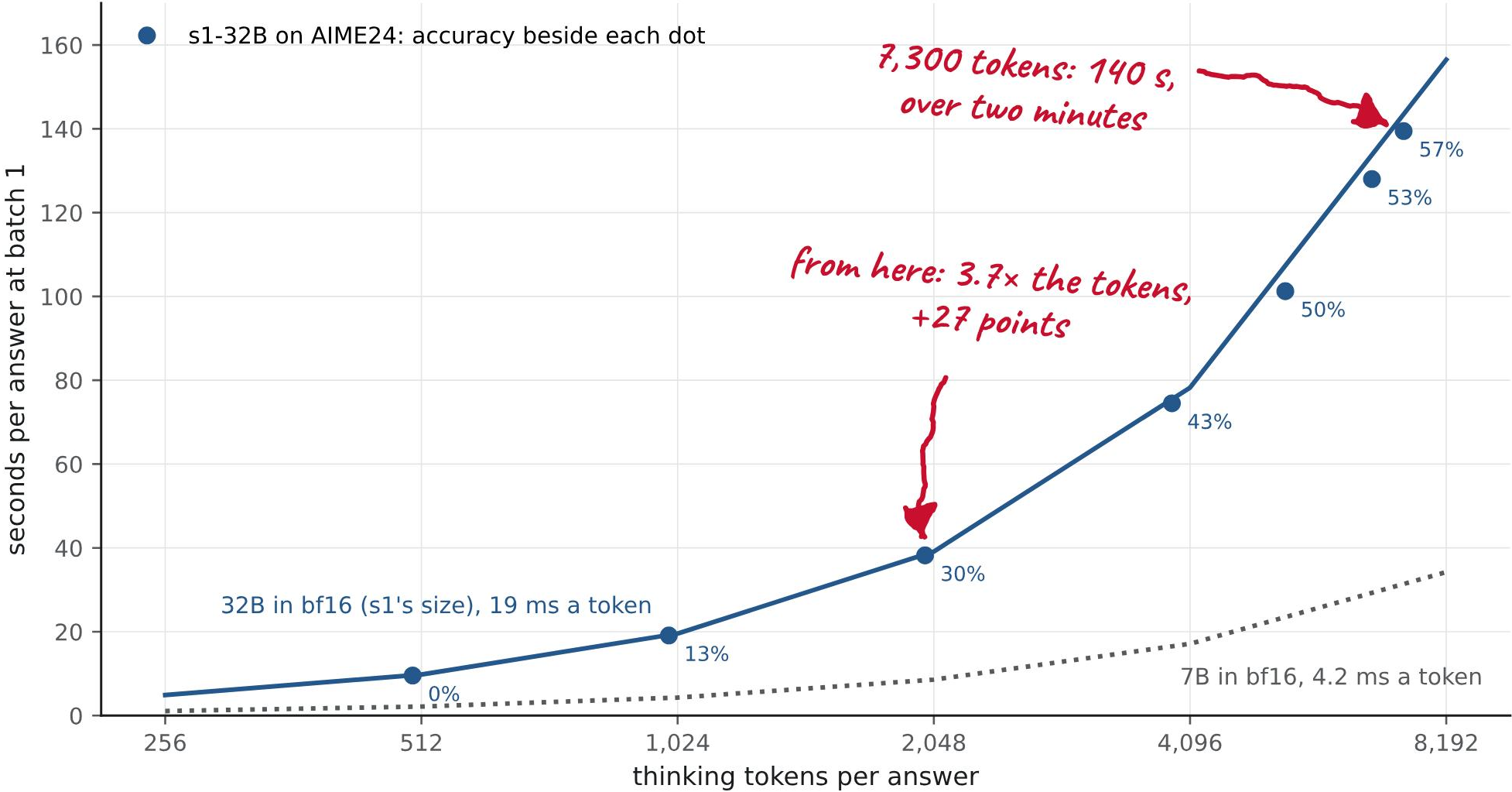
Response

*set per question,
no retraining*

*it tried to stop here:
end token suppressed,
"Wait" appended*

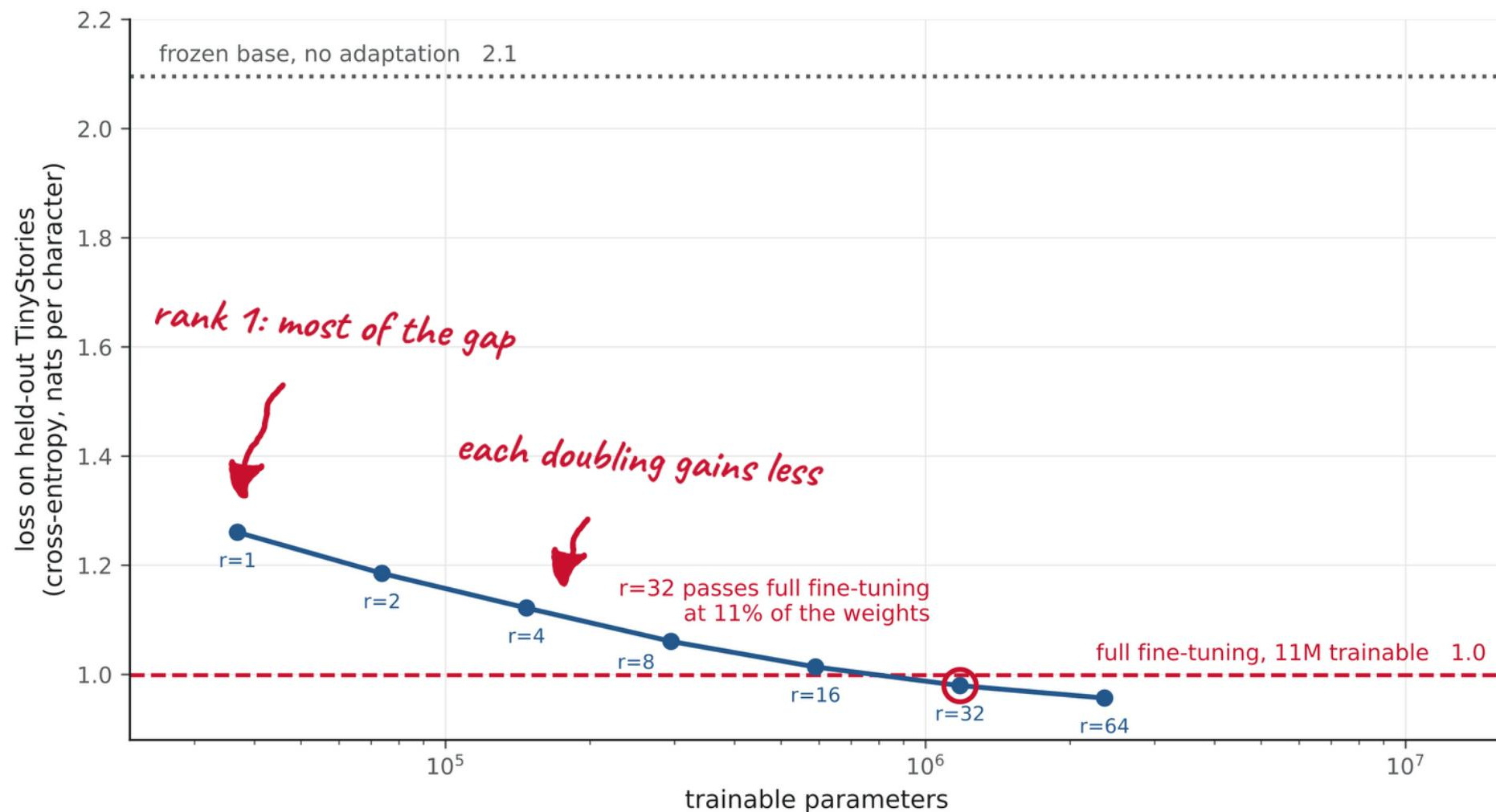
*forcing less: at the limit,
insert the end token*

Thinking as Time



Time per token from the decode arithmetic in section one: weight bytes over 3.35 TB/s on one H100 at batch 1. s1-32B thinking lengths and AIME24 accuracies from Muennighoff et al., 2025, Figure 1.

Assignment 5



Reading

Required

Chen, Zaharia and Zou, FrugalGPT · [arXiv:2305.05176](#)

Optional

Muennighoff et al., s1 · [arXiv:2501.19393](#) Ong et al., RouteLLM · [arXiv:2406.18665](#)

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Assignment 5, out today, due October 20 with its quiz. The project concept document is due October 13.